Name

Mathematics Advanced Year 12

Differentiating Trigonometric Functions

Book 3 Differentiation with trigonometric functions

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Contents

Syllabus Content 2
Learning Intention ..... **Error! Bookmark not defined.**

Syllabus Content

MAV-12-04 selects and applies differentiation methods to solve problems

Differentiation with trigonometric functions

- Determine the formulas $\frac{d}{dx}(\sin x) = \cos x$ and $\frac{d}{dx}(\cos x) = -\sin x$ informally by using the graphs of $y = \sin x$ and $y = \cos x$ and verify with graphing applications
- Use the rules of differentiation to show that $\frac{d}{dx}(\tan x) = \sec^2 x$
- Use the rules of differentiation to find the derivatives of $\operatorname{cosec} x$, $\sec x$ and $\cot x$
- Use the chain rule to differentiate functions of the form $\sin f(x)$, $\cos f(x)$ and $\tan f(x)$

Derivative of the Trigonometric Functions

Derivative of $\sin x$ and $\cos x$

https://www.geogebra.org/m/xqmt2tfk

$y = \sin x$
$y' = \cos x$

$y = \cos x$
$y' = -\sin x$

Differentiate $\sin x$ and $\cos x$.

$y = 3\sin x$
$y' = 3\cos x$

$y = 5\cos x$
$y' = -5\sin x$

$y = x^3 + 2x - \sin x$
$y' = 3x^2 + 2 - \cos x$

Differentiate these functions.

a $f(x) = 5\cos x$

b $f(x) = 5\sin x$

c $f(x) = 3 - 2\sin x$

d $f(x) = -9\cos x$

e $f(x) = 2x^4 - \frac{\sin x}{5}$

f $f(x) = \sin x - \cos x$

Investigation Derivative of $\tan x$.

By rewriting $\tan x = \frac{\sin x}{\cos x}$ and using quotient rule, find the derivative of $\tan x$.

$y = \frac{\sin x}{\cos x}$

$y' = \frac{u'v - uv'}{v^2}$

Let $u = \sin x$ $v = \cos x$

$u' =$ $v' =$

ⓘ Derivative of $\tan x$

$$y = \tan x$$
$$y' = \sec^2 x$$

⚠ $\sec^2 x = (\sec x)^2$

Differentiate $\tan x$

Differentiate these functions.

a	$f(x) = 7\cos x$	**b**	$f(x) = -7\sin x$	**c**	$f(x) = 4 - \tan x$
d	$f(x) = 4x^2 - 13\tan x$	**e**	$f(x) = \frac{\tan x}{51} + 3e^x$	**f**	$f(x) = \cos x + \tan x$

ⓘ Chain Rule for Trigonometric Functions

$y = \sin f(x)$ $y' = f'(x)\cos f(x)$	$y = \cos f(x)$ $y' = -f'(x)\sin(x)$	$y = \tan f(x)$ $y' = f'(x)\sec^2(x)$

Differentiate trigonometric functions using the chain rule

Differentiate these functions.

a	$f(x) = \sin 2x$	**b**	$f(x) = \cos 2x$	**c**	$f(x) = \tan 7x$
d	$f(x) = 5\cos(8x + 1)$	**e**	$f(x) = \sin(x^2)$	**f**	$f(x) = -5\tan(x^3 + 2x)$

Example Differentiate trigonometric functions.

Differentiate

$y = \sin(3x)\cos(5x)$ use product rule

$y' = u'v + uv'$

$= (3\cos(3x))(\cos(5x)) + (\sin(3x))(-5\sin(5x))$

$= 3\cos(3x)\cos(5x) - 5\sin(3x)\sin(5x)$

Differentiate

$y = \sin^4(2x)$ use chain rule

$= (\sin(2x))^4$

$\frac{dy}{dx} = 4(\sin(2x))^3 \times 2\cos(2x)$

$= 8\cos 2x\,(\sin(2x))^3$

$= 8\cos 2x \sin^3(2x)$

Practice

Differentiate

$y = \sin\left(x^2 - \frac{\pi}{4}\right)$

$2x\cos\left(x^2 - \frac{\pi}{4}\right)$

Differentiate

$y = \tan(e^{2x})$

$2e^{2x}\sec^2(e^{2x})$

Differentiate

$y = \cos^2(3x + \pi)$

$6\cos(3x + \pi)\sin(3x + \pi)$

Differentiate

$y = \frac{\sin 2x}{x}$

$\frac{2x\cos(2x) - \sin(2x)}{x^2}$

1 Differentiate with respect to x.

a $y = \sin x$

$\frac{dy}{dx} = \cos x$

b $y = \cos x$

$\frac{dy}{dx} = -\sin x$

c $y = \tan x$

$\frac{dy}{dx} = \sec^2 x$

d $y = 2\sin x$

$\frac{dy}{dx} = 2\cos x$

e $y = \sin 2x$

$\frac{dy}{dx} = 2\cos 2x$

f $y = 3\cos x$

$\frac{dy}{dx} = -3\sin x$

g $y = \cos 3x$

$\frac{dy}{dx} = -3\sin 3x$

h $y = \tan 4x$

$\frac{dy}{dx} = 4\sec^2 4x$

i $y = 4\tan x$

$\frac{dy}{dx} = 4\sec^2 x$

j $y = 2\sin 3x$

$\frac{dy}{dx} = 6\cos 3x$

k $y = 2\tan 2x$

$\frac{dy}{dx} = 4\sec^2 2x$

l $y = 4\cos 2x$

$\frac{dy}{dx} = -8\sin 2x$

m $y = -\sin 2x$

$\frac{dy}{dx} = -2\cos 2x$

n $y = -\cos 2x$

$dy/dx = 2\sin 2x$

o $y = -\tan 2x$

$\frac{dy}{dx} = -2\sec^2 2x$

p $y = \tan\left(\frac{1}{2}x\right)$

$\frac{dy}{dx} = \frac{1}{2}\sec^2\left(\frac{x}{2}\right)$

q $y = \cos\left(\frac{1}{2}x\right)$

$\frac{dy}{dx} = -\frac{1}{2}\sin\left(\frac{x}{2}\right)$

r $y = \sin\left(\frac{x}{3}\right)$

$\frac{dy}{dx} = \frac{1}{3}\cos\left(\frac{x}{3}\right)$

s $y = 5\tan\left(\frac{1}{5}x\right)$

$\frac{dy}{dx} = \sec^2\left(\frac{x}{5}\right)$

t $y = 6\cos\left(\frac{x}{3}\right)$

$\frac{dy}{dx} = -2\sin\left(\frac{x}{3}\right)$

u $y = 12\sin\left(\frac{x}{4}\right)$

$\frac{dy}{dx} = 3\cos\left(\frac{x}{4}\right)$

2 Differentiate with respect to x.

a $\sin 2\pi x$

$\frac{dy}{dx} = 2\pi\cos 2\pi x$

b $\tan\left(\frac{\pi x}{2}\right)$

$\frac{dy}{dx} = \frac{\pi}{2}\sec^2\left(\frac{\pi x}{2}\right)$

c $3\sin x + \cos 5x$

$\frac{dy}{dx} = 3\cos x - 5\sin 5x$

d $4\sin \pi x + 3\cos \pi x$

$\frac{dy}{dx} = 4\pi\cos \pi x - 3\pi\sin \pi x$

e $\sin(2x - 1)$

$\frac{dy}{dx} = 2\cos(2x - 1)$

f $\tan(1 + 3x)$

$dy/dx = 3\sec^2(1 + 3x)$

g $2\cos(1 - x)$

$\frac{dy}{dx} = 2\sin(1 - x)$

h $\cos(5x + 4)$

$\frac{dy}{dx} = -5\sin(5x + 4)$

i $7\sin(2 - 3x)$

$\frac{dy}{dx} = -21\cos(2 - 3x)$

j $10\tan(10 - x)$

$\frac{dy}{dx} = -10\sec^2(10 - x)$

k $6\sin\left(\frac{x+1}{2}\right)$

$\frac{dy}{dx} = 3\cos\left(\frac{x+1}{2}\right)$

l $15\cos\left(\frac{2x+1}{5}\right)$

$\frac{dy}{dx} = -6\sin\left(\frac{2x+1}{5}\right)$

3 Find the first, second, third and fourth derivatives of:

a $y = \sin 2x$

$y = \sin 2x$
$y' = 2\cos 2x$
$y'' = -4\sin 2x$
$y''' = -8\cos 2x$
$y'''' = 16\sin 2x$

b $y = \cos 10x$

$y = \cos 10x$
$y' = -10\sin 10x$
$y'' = -100\cos 10x$
$y''' = 1000\sin 10x$
$y'''' = 10000\cos 10x$

c $y = \sin\left(\frac{x}{2}\right)$

$y = \sin\left(\frac{x}{2}\right)$
$y' = \left(\frac{1}{2}\right)\cos\left(\frac{x}{2}\right)$
$y'' = -\left(\frac{1}{4}\right)\sin\left(\frac{x}{2}\right)$
$y''' = -\left(\frac{1}{8}\right)\cos\left(\frac{x}{2}\right)$
$y'''' = \left(\frac{1}{16}\right)\sin\left(\frac{x}{2}\right)$

d $y = \cos\left(\frac{x}{3}\right)$

$y = \cos\left(\frac{x}{3}\right)$
$y' = -\left(\frac{1}{3}\right)\sin\left(\frac{x}{3}\right)$
$y'' = -\left(\frac{1}{9}\right)\cos\left(\frac{x}{3}\right)$
$y''' = \left(\frac{1}{27}\right)\sin\left(\frac{x}{3}\right)$
$y'''' = \left(\frac{1}{81}\right)\cos\left(\frac{x}{3}\right)$

4 If $f(x) = \cos 2x$, find $f'(x)$ and then find:

$f'(x) = -2\sin 2x$

a $f'(0)$

$f'(0) = -2\sin 0 = 0$

b $f'\left(\frac{\pi}{12}\right)$

$f'\left(\frac{\pi}{12}\right) = -2\sin\left(\frac{\pi}{6}\right) = -2\left(\frac{1}{2}\right) = -1$

c $f'\left(\frac{\pi}{6}\right)$

$f'\left(\frac{\pi}{6}\right) = -2\sin\left(\frac{\pi}{3}\right) = -2\left(\frac{\sqrt{3}}{2}\right) = -\sqrt{3}$

d $f'\left(\frac{\pi}{4}\right)$

$f'\left(\frac{\pi}{4}\right) = -2\sin\left(\frac{\pi}{2}\right) = -2(1) = -2$

5 Find the equation of the tangent to $y = \sin(\pi - x)$ at the point $\left(\frac{\pi}{6}, \frac{1}{2}\right)$ in exact general form.

$\frac{dy}{dx} = -\cos(\pi - x)$

$At\ x = \frac{\pi}{6}: \frac{dy}{dx} = -\cos\left(\pi - \frac{\pi}{6}\right) = -\cos\left(\frac{5\pi}{6}\right) = \frac{\sqrt{3}}{2}$

Tangent: $y - \frac{1}{2} = \left(\frac{\sqrt{3}}{2}\right)\left(x - \frac{\pi}{6}\right)$

This simplifies to

$6\sqrt{3}x - 12y + 6 - \pi\sqrt{3} = 0$

6 Find the equation of the normal to $y = 3\sin(2x)$ at the point where $x = \frac{\pi}{8}$ in exact general form.

$\frac{dy}{dx} = 6\cos(2x)$

$At\ x = \frac{\pi}{8}: \frac{dy}{dx} = 6\cos\left(\frac{\pi}{4}\right) = 6\left(\frac{\sqrt{2}}{2}\right) = 3\sqrt{2}$

Normal gradient $= -\frac{1}{3\sqrt{2}} = -\frac{\sqrt{2}}{6}$

$At\ x = \frac{\pi}{8}: y = 3\sin\left(\frac{\pi}{4}\right) = \frac{3\sqrt{2}}{2}$

Normal: $y - \frac{3\sqrt{2}}{2} = \left(-\frac{\sqrt{2}}{6}\right)\left(x - \frac{\pi}{8}\right)$

This simplifies to

$8x + 24\sqrt{2}y - 72 - \pi = 0$

7 Find $\frac{dy}{dx}$ using the product rule.

a $y = x \sin x$

$u = x, u' = 1;\ v = \sin x, v' = \cos x$
$\frac{dy}{dx} = x\cos x + \sin x$

b $y = 2x \tan 2x$

$u = 2x, u' = 2;\ v = \tan 2x, v' = 2\sec^2 2x$
$\frac{dy}{dx} = 2(\tan 2x + 2x\sec^2 2x)$

c $y = x^2 \cos 2x$

$u = x^2, u' = 2x;\ v = \cos 2x, v' = -2\sin 2x$
$\frac{dy}{dx} = 2x(\cos 2x - x\sin 2x)$

d $y = x^3 \sin 3x$

$u = x^3, u' = 3x^2;\ v = \sin 3x, v' = 3\cos 3x$
$\frac{dy}{dx} = 3x^2(\sin 3x + x\cos 3x)$

8 Find $\frac{dy}{dx}$ using the quotient rule.

a $y = \frac{\sin x}{x}$

$u = \sin x, u' = \cos x;\ v = x, v' = 1$
$\frac{dy}{dx} = \frac{x\cos x - \sin x}{x^2}$

b $y = \frac{\cos x}{x}$

$u = \cos x, u' = -\sin x;\ v = x, v' = 1$
$\frac{dy}{dx} = \frac{-x\sin x - \cos x}{x^2}$

c $y = \frac{x^2}{\cos x}$

$u = x^2, u' = 2x;\ v = \cos x, v' = -\sin x$
$\frac{dy}{dx} = \frac{2x\cos x + x^2\sin x}{\cos^2 x}$
$= \frac{x(2\cos x + x\sin x)}{\cos^2 x}$

d $y = \frac{x}{1 + \sin x}$

$u = x, u' = 1;\ v = 1 + \sin x, v' = \cos x$
$\frac{dy}{dx} = \frac{1 + \sin x - x\cos x}{(1 + \sin x)^2}$

9 Find $\frac{dy}{dx}$ using the chain rule. Remember that $\cos^2 x$ means $(\cos x)^2$.

a $y = \sin(x^2)$

$\frac{dy}{dx} = 2x\cos(x^2)$

b $y = \sin(1 - x^2)$

$\frac{dy}{dx} = -2x\cos(1 - x^2)$

c $y = \cos(x^3 + 1)$

$\frac{dy}{dx} = -3x^2\sin(x^3 + 1)$

d $y = \sin\left(\frac{1}{x}\right)$

$\frac{dy}{dx} = -\left(\frac{1}{x^2}\right)\cos\left(\frac{1}{x}\right)$

e $y = \cos^2 x$

$\frac{dy}{dx} = -2\cos x \sin x$

f $y = \sin^3 x$

$\frac{dy}{dx} = 3\sin^2 x \cos x$

g $y = \tan^2 x$

$\frac{dy}{dx} = 2\tan x \sec^2 x$

h $y = \tan\sqrt{x}$

$dy/dx = \frac{1}{2\sqrt{x}}\sec^2\sqrt{x}$

10 Differentiate these functions.

a $y = \sin x \cos x$

$\frac{dy}{dx} = \cos^2 x - \sin^2 x$

b $y = \sin^2 7x$

$\frac{dy}{dx} = 2\sin 7x\,(7\cos 7x)$
$= 14\sin 7x \cos 7x$

c $y = \cos^5 3x$

$\frac{dy}{dx} = 5\cos^4 3x\,(-3\sin 3x)$
$= -15\cos^4 3x \sin 3x$

d $y = (1 - \cos 3x)^3$

$\frac{dy}{dx} = 3(1 - \cos 3x)^2(3\sin 3x)$
$= 9\sin 3x\,(1 - \cos 3x)^2$

e $y = \sin 2x \sin 4x$

$\frac{dy}{dx} = 2\cos 2x \sin 4x + 4\sin 2x \cos 4x$
$= 2(\cos 2x \sin 4x + 2\sin 2x \cos 4x)$

f $y = \tan^3(5x - 4)$

$\frac{dy}{dx} = 3\tan^2(5x - 4)\left(5\sec^2(5x - 4)\right)$
$= 15\tan^2(5x - 4)\sec^2(5x - 4)$

11 Find $f'(x)$, given that:

a $f(x) = \frac{1}{1 + \sin x}$

$\frac{-\cos x}{(1 + \sin x)^2}$

b $f(x) = \frac{\sin x}{1 + \cos x}$

$\frac{1}{1 + \cos x}$

c $f(x) = \frac{1 - \sin x}{\cos x}$

$= -\frac{1}{1 + \sin x}$

d $f(x) = \frac{\cos x}{\cos x + \sin x}$

$= -\frac{1}{(\cos x + \sin x)^2}$

12 Consider the function $y = \frac{\tan^3 x}{3} - \tan x + x$.

a Show that $\frac{dy}{dx} = \tan^2 x \sec^2 x - \sec^2 x + 1$.

$\frac{dy}{dx} = \left(\frac{1}{3}\right)(3\tan^2 x \sec^2 x) - \sec^2 x + 1$
$= \tan^2 x \sec^2 x - \sec^2 x + 1$

b Hence use the identity $\sec^2 x = 1 + \tan^2 x$ to show that $\frac{dy}{dx} = \tan^4 x$.

$\frac{dy}{dx} = \tan^2 x\,(1 + \tan^2 x) - (1 + \tan^2 x) + 1$
$= \tan^2 x + \tan^4 x - 1 - \tan^2 x + 1$
$= \tan^4 x$

13

a By writing $\sec x$ as $(\cos x)^{-1}$, show that $\frac{d}{dx}(\sec x) = \sec x \tan x$.

$\frac{d}{dx}(sec\ x) = -(cos\ x)^{-2}(-sin\ x)$

$= \frac{\sin x}{\cos^2 x}$

$= \left(\frac{1}{\cos x}\right)\left(\frac{\sin x}{\cos x}\right)$

$= \sec x \tan x$

b Similarly, show that $\frac{d}{dx}(\text{cosec}\ x) = -\text{cosec}\ x \cot x$.

$\frac{d}{dx}(cosec\ x) = -(sin\ x)^{-2}(cos\ x)$

$= -\frac{\cos x}{\sin^2 x}$

$= -\left(\frac{1}{\sin x}\right)\left(\frac{\sin x}{\cos x}\right)$

$= -cosec\ x\ cot\ x$

c Similarly, show that $\frac{d}{dx}(\cot x) = -\text{cosec}^2 x$.

$\frac{d}{dx}(cot\ x) = \frac{sin\ x\ (-sin\ x) - cos\ x\ (cos\ x)}{sin^2 x}$

$= \frac{-sin^2 x - cos^2 x}{sin^2 x}$

$= -\frac{1}{sin^2 x}$

$= -cosec^2 x$

14 Show that $\frac{d}{dx}\left(\frac{\sin^5 x}{5} - \frac{\sin^7 x}{7}\right) = \sin^4 x \cos^3 x$.

$\frac{dy}{dx} = \left(\frac{1}{5}\right)(5\ sin^4 x\ cos\ x) - \left(\frac{1}{7}\right)(7\ sin^6 x\ cos\ x)$

$= sin^4 x \cos x - sin^6 x \cos x$

$= sin^4 x \cos x\ (1 - sin^2 x)$

$= sin^4 x \cos x\ cos^2 x$

$= sin^4 x\ cos^3 x$

15 Show that:

a $\frac{d}{dx}\left(\cos^2(3x + \pi)\right) = -6 \sin 3x \cos 3x$

$\frac{dy}{dx} = 2\ cos(3x + \pi)\left(-3\ sin(3x + \pi)\right) = -6\ sin(3x + \pi)cos(3x + \pi)$

Using $\sin(\pi + \theta) = -\sin\theta$ *and* $\cos(\pi + \theta) = -\cos\theta$

$\frac{dy}{dx} = -6(-sin\ 3x)(-cos\ 3x)$

$= -6\ sin\ 3x\ cos\ 3x$

b $\frac{d}{dx}\left(\sin\left(\frac{\pi}{2} - x^2\right)\right) = -2x \sin(x^2)$

$\frac{dy}{dx} = -2x \cos\left(\frac{\pi}{2} - x^2\right)$

Using $\cos\left(\frac{\pi}{2} - \theta\right) = \sin(\theta)$

$\frac{dy}{dx} = -2x \sin(x^2)$